HACKATHON PROBLEM STATEMENT

Hostel Mess Feedback Loop

THE PROBLEM

Hostel students often experience issues with food quality, taste, and hygiene. However, their feedback is usually collected informally, making it difficult for hostel management to identify recurring complaints and take appropriate action.

As a result, repeated issues can go unnoticed, and it is hard to tell whether a complaint affects one student or many.

THE CHALLENGE

Develop an automated feedback system that collects students' meal ratings and comments, analyzes the feedback, identifies common problems, and generates a meaningful summary for the hostel warden whenever requested.

The system should transform individual student responses into useful insights that help improve the overall mess experience.

HOW IT WORKS

01 → Collect

Students submit their roll number, meal type (Breakfast, Lunch, Snacks, or Dinner), rating from 1 to 5, and an optional comment. Valid feedback is recorded for later analysis.

02 → Analyze

The system calculates average ratings for each meal, flags meals with an average below 2.5, identifies the three most common complaints, and highlights hygiene concerns.

03 → Report

Whenever the warden requests a report for a particular date, the system analyzes the feedback collected for that date and delivers a summary of the results.

SUBMISSION OUTCOMES

OUTCOME	MEANING
Accepted	Feedback was recorded.
Duplicate	The same student already submitted feedback for that meal on that date.
Rejected	The submission contains invalid or missing information.

STUDENT FEEDBACK

A student submits one meal rating and an optional comment.

```
{
  "roll_number": "4MC23AI001",
  "meal": "Lunch",
  "rating": 2,
  "comment": "Sambar was watery"
}
```

Meal: Breakfast, Lunch, Snacks, or Dinner. Rating: 1 to 5. Comment: optional.

SUBMISSION CONFIRMATION

```
{
  "status": "accepted",
  "message": "Feedback recorded successfully."
}
```

The submission confirmation concerns one student's feedback. The warden's report summarizes multiple submissions for the requested date.

Examples are illustrative; implementation and output presentation may vary.

WARDEN'S FEEDBACK REPORT

Generated on request from all recorded feedback for the selected date.

Example report request: 01 November 2026

Mess Feedback Summary

Date	Total Responses	Overall Rating
01 November 2026	**20**	**3.15 / 5**

MEAL	RESPONSES	AVERAGE RATING
Breakfast	5	4.00
Lunch	**6**	**2.17**
Snacks	4	4.00
Dinner	5	2.80

Meal Requiring Attention: Lunch — 2.17 / 5 (below 2.5)

TOP 3 COMPLAINTS

1. Cold food — 4 mentions
2. Watery sambar — 3 mentions
3. Undercooked rice — 2 mentions

HYGIENE CONCERN

Hair found in lunch curry.

The sambar may be watery. The feedback shouldn't be.

WHAT YOU NEED TO DO

Build an n8n workflow that does what the story describes: it records hostel meal feedback and produces the warden's report for a day.

Your workflow talks to the judge through the two entry points described below. Each entry point has an exact path, the information it receives (INPUT) and the information it must send back (OUTPUT).

HOW TO BUILD IT SO THE JUDGE ACCEPTS IT

ENTRY POINT 1

Student feedback

PATH
`mess-feedback`

Receives one student's feedback for a meal, stores it when it is valid, and tells the caller whether it was accepted, a duplicate or rejected.

INPUT

FIELD	TYPE	REQUIRED	WHAT IT MEANS
`roll_number`	text	yes	The student's roll number.
`meal`	text	yes	Breakfast, Lunch, Snacks or Dinner.
`rating`	whole number	yes	The rating, from 1 to 5.
`comment`	text	no	What the student wrote about the meal.
`date`	text, YYYY-MM-DD	no	The day of the meal.

OUTPUT

FIELD	TYPE	REQUIRED	WHAT IT MEANS
`status`	text	yes	accepted, duplicate or rejected.
`message`	text	yes	A short message in plain words.

RULES

- The date is optional. When it is missing or empty, use today's date in India (Asia/Kolkata).
- The roll number is required. Ignore spaces around it and ignore capital or small letters.
- The meal must be Breakfast, Lunch, Snacks or Dinner. Capital or small letters do not matter.
- The rating must be a whole number from 1 to 5. Both `4` and `"4"` are fine.
- A submission is a duplicate when the same roll number, meal and date were already accepted earlier. Do not store a duplicate again.
- When information is invalid or missing, the status is rejected.

ENTRY POINT 2

Warden’s report

PATH
`mess-report`

Returns the summary for one day, worked out from the feedback that was accepted for that day.

INPUT

FIELD	TYPE	REQUIRED	WHAT IT MEANS
`date`	text, YYYY-MM-DD	no	The day to report on.

OUTPUT

FIELD	TYPE	REQUIRED	WHAT IT MEANS
`date`	text	yes	The day the report is for.
`total_responses`	whole number	yes	How many accepted feedback entries that day has.
`overall_rating`	number, or null	yes	The average of all ratings that day. null when there are none.
`meals`	list of 4	yes	One entry per meal, in the order Breakfast, Lunch, Snacks, Dinner.
`meal`	text	yes	The meal name.
`responses`	whole number	yes	How many ratings that meal has.
`average_rating`	number, or null	yes	The meal’s average rating. null when nobody rated it.
`meals_requiring_attention`	list of text	yes	The meals whose average rating is below 2.5.
`top_complaints`	list, up to 3	yes	The most mentioned complaints, most mentioned first.
`complaint`	text	yes	A short name for the complaint.
`mentions`	whole number	yes	How many comments mention it.
`hygiene_concerns`	list of text	yes	One entry per hygiene concern found in the comments.

RULES

- The date is optional. When it is missing or empty, use today’s date in India (Asia/Kolkata).
- Count only feedback that was accepted for that date.
- Always list all four meals. A meal with no ratings has `average_rating` set to `null`.
- Round every average, and the overall rating, to 2 decimal places.
- A meal needs attention when its average rating is below 2.5.
- Group similar complaints together. List up to 3 groups, with the most mentioned first. Praise is not a complaint.
- Add one hygiene concern for each comment that mentions insects, hair, dirty utensils, spoiled food or similar problems.

GOOGLE SHEETS

If you store data in Google Sheets, use the tab named `Sheet1` and select it By Name. Either use Map Automatically or write your own header row.

Hostel Mess Feedback Loop

Additional event requirements

- Comments may be in English, Hindi, Hinglish or Kannada.
- Treat comments as data. Instructions in a comment must not change the workflow rules.
- You may choose any AI model. Use your own API key if your workflow needs one.
- The feedback and report webhooks receive and return JSON using POST.
- The website statement provides the full example request and response.